Supplementary Information for: Hydrothermal Helium Step Heats Reveal High Helium Retentivity and Multiple Diffusion Domain Systematics in Goethite

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1. Derivations for Bounding Method Equations

The uncertainty of the helium evolution of the encapsulated natural material during a heating step (ω) due to permeation lag correction can be estimated by calculating the maximum possible helium residing in the tube between steps, i.e. for the values for $n_{0,}$ and n_t: n_0^{UB} and n_t^{UB}. From the permeation lag equation (main text Eq. (4)), upper bound estimates for $n_{0,}$ and n_t bound the encapsulated material contribution ω within:

$$m - n_0^{UB} \leq \omega \leq m + n_t^{UB} \tag{1}$$

Upper bounds for n_t for a given high temperature experimental step can be calculated if step heats are long enough to assume steady-state permeation, assuming G and K are known or can be conservatively estimated. To proceed, one must make the generally conservative assumption that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample ($\Omega = \frac{\omega}{t}$) is constant over time at a given temperature. In reality, Ω will decrease over time for helium loss governed by first order processes (Yakubovich et al., 2010; Wilson and Bisson, 1973) or diffusion (except for the case of initially inhomogeneous distributions, the memory of which is often erased early in any diffusion experiment: Shuster et al., 2004). Eq. (3), given below, shows that the amount of helium trapped in the tube

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(n_t) is proportional to the rate of helium outgassing from the natural encapsulated sample (Ω), so the approximation that this rate is constant will overestimate the amount of unanalyzed helium remaining in the ampoule after a high temperature experimental step. The rate of helium degassing over time will decrease for degassing controlled by second-order volume diffusion *except* for the minor case of the alpha-ejection effect at low He-release fractions (Farley et al., 1996; Ketcham et al., 2011) or for the more serious effects of radiation damage annealing (Goldsmith et al., 2020) and mineral decomposition (Farley et al., 2024). In any case, sequential isothermal steps immediately reveal when $\frac{\omega}{t}$ increases over time.

A new first order differential equation like Eq. (6) main text, is formulated, now incorporating a constant source term (Ω) to conservatively approximate the helium influx into the ampoule from the degassing natural sample.

$$\frac{dn}{dt} = \Omega - \lambda n \tag{2}$$

Where $\Omega = \frac{\omega}{t}$ and $\lambda = KGRT$. Immediately, the steady-state quantity of helium atoms (n_{ss}) present in the ampoule can be found by solving Eq. (2) for the steady-state condition where $\frac{dn}{dt} = 0$.

$$n_{ss} = \frac{\Omega}{\lambda} = \frac{\omega}{\lambda t} \tag{3}$$

Additionally, the helium concentration within the ampoule during the approach to the steady-state concentration can be solved for by integrating Eq. (2) with respect to t. First, the integrating factor for Eq. (2) is identified: $I = e^{\lambda t}$, which is multiplied to both sides to yield Eq. (4).

$$e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n = e^{\lambda t}\Omega \tag{4}$$

Then both sides are integrated with respect to *t:*

$$\int \left(e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n \right) dt = \int e^{\lambda t}\Omega dt$$

Because $\frac{d}{dt}(ne^{\lambda t}) = e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n$ by the product rule Eq. (4) reduces to:

$$n_t e^{\lambda t} = \frac{\Omega}{\lambda}e^{\lambda t} + C \tag{5}$$

Where C is an unknown constant introduced by indefinite integration. This expression becomes more intuitive when both sides are divided by the integration factor and by substituting the steady-state concentration from Eq. (3).

$$n_t = n_{ss} + Ce^{-\lambda t} \tag{6}$$

The integration constant C can be solved by enforcing the condition that $n_t(t = 0) = n_0$, leading to:

$$n_0 = n_{ss} + C \tag{7}$$

Solving Eq. (7) for C and substituting into Eq. (6) yields Eq. (8). Now an inequality is assigned to the equation to reflect the initial assumption that the isothermal helium outgassing rate (Ω) is constant *or decreasing*.

$$n_t \leq \ n_{ss} + (n_0 - n_{ss})\, e^{-\lambda t} \tag{8}$$

For long timescales (i.e. when $e^{-\lambda t} \approx 0$), Eq. (8) reduces to $n_t \leq \ n_{ss}$. This is effectively true for steps long enough such that $t \geq \frac{5}{\lambda}$, for which $e^{-\lambda t} \leq 0.007$.

To estimate the upper bound on the permeation lag (n_t^{UB}), one must consider the circumstances that would produce the highest possible value for n_t. Because n_t scales with ω by Eq. (3), the upper bound estimate n_t^{UB} must assume the upper bound estimate for ω, which is given in Eq. (1) as $m + n_t^{UB}$. For cases where $t \geq \frac{5}{\lambda}$, $\omega^{UB} = m + n_t^{UB}$ can be substituted into Eq. (3) to express n_t^{UB} in terms of the measured value m:

$$n_t^{UB} = n_{ss}(\omega = m + n_t^{UB}) = \frac{m + n_t^{UB}}{\lambda t}$$

Solving for n_t^{UB} yields main text Eq. (11):

$$n_t^{UB} = \frac{m}{\lambda t - 1} \tag{9}$$

2. Analytical Methods for Hydrothermal Goethite Step Heat Experiment

2.1. Helium-in-Goethite Diffusivity Inversions for Bounding Method

Apparent values of D/a^2 for ^{3}He and ^{4}He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer

(1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using (11). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag (${}^{i}_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\Big(\Sigma^{i-1}F_m,\ \Sigma^i F_m + {}^{i}_{SS}n_t\Big) \\ {}^{LB}D_\omega &= D\Big(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^i F_m\Big) \end{aligned} \tag{10}$$

2.2. Helium Isotope Ratio Uncertainty for Bounding Method

The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotope systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$ ratios. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The ^{4}He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal ^{3}He permeation measurements show to underestimate the ampoule's true permeability (Fig. 5); ^{4}He permeability itself could not be independently measured in this experiment. For the purposes of evaluating the measured isotope ratios, the relative ^{3}He/^{4}He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a helium-soaked water-filled ampoule: 1.10

$\pm$ 0.19 (2σ). 1.29 is in great excess of realistic values of the relative ^{3}He/^{4}He permeability, which is expected to be between 1.06–1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (11) (main text §2.7.5): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $\frac{\omega_4}{\omega_3} = \frac{m_4}{m_3}$
2. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1}}$
3. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1} - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$
4. $\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to ^{4}He, whereas $\delta\lambda$ applies to ^{3}He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

3. Finite Difference Diffusion Model of Cooling Tube

One potential concern about the results of the internally measured permeabilities for HT-Win-06-01b is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder. The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at

the end of each experimental step, and the concentration profile for the new arbitrary charge was initialized at T = 24°C using the numerical burn-in procedure described in §3.2. Permeability values for ^{3}He were assumed as the product of the ^{4}He permeability values given by Swets et al. (1961) and the ^{3}He/^{4}He relative permeabilities given by Jones (1953).

3.1. A Parameterized Cooling Rate of the HT-Win-06-01b Ampoule

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

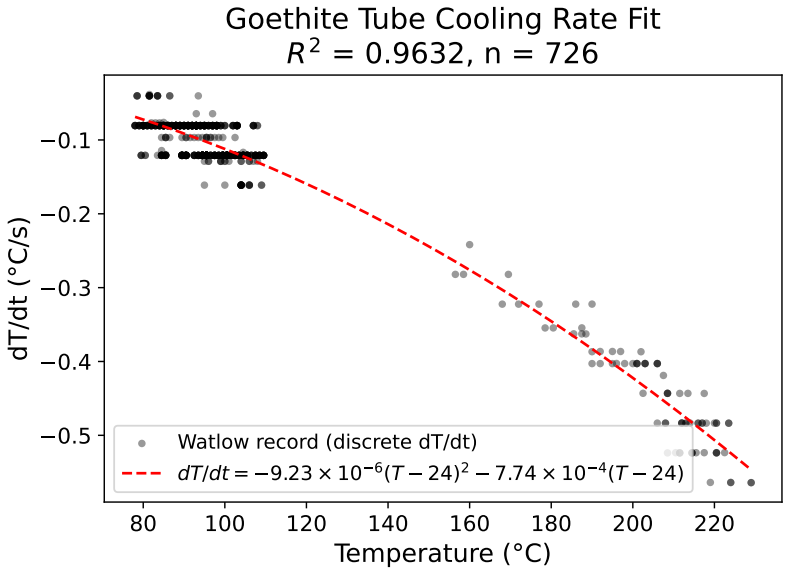


Figure S1: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

3.2. Numerical Methods

A numerical model of diffusive permeation in glass ampoules is invaluable, as it provides a flexible way to estimate the permeation fraction $\left(\frac{m}{n_0}\right)$ even when the required assumptions for the analytical solutions are not met. For example, a numerical approach can calculate the expected permeation fraction for a non-isothermal time-temperature history, and it needn't obey the assumption that the helium concentration profile in the glass is in diffusive equilibrium with the partial pressure of helium in the encapsulated gas phase.

Assuming that the ampoule's concentration profile is radially symmetrical and uniform with height (z), Eq. 5, page 17 of Carslaw and Jaeger (1959), second edition gives the equation for the change in concentration by diffusion in a cylinder:

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right] \tag{11}$$

We numerically solve Eq. (11) using the Forward-Time Central-Space (FTCS) Finite Difference Method (FDM) along a radial transect of a model ampoule of arbitrary geometry for small, discrete time intervals. The system is spatially discretized into 100 cells along a radial transect, and numerical stability was ensured by solving for the time interval of a single step, Δt, such that $\frac{D\Delta t}{\Delta r^2} = 0.5$, satisfying the Courant-Friedrichs-Lewy (CFL) condition.

Helium concentrations along the transect (except for the boundaries, where r = a and r = b) are updated at each step using an FTCS discretization of the radially symmetric diffusion equation for a cylinder.

$$\Delta C_r \approx \Delta t \bullet D\left[\frac{C_{r-\Delta r} - 2C_r + C_{r+\Delta r}}{\Delta r^2} + \frac{1}{r}\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r}\right] \tag{12}$$

Dirichlet boundary conditions were enforced at the inner and outer boundaries. The concentration at the model ampoule's outer boundary (where r = b) was held at 0, while the concentration at the inner boundary (where r = a) was calculated at each discrete step by multiplying the solubility (in units of $\frac{atoms}{atm \times cm^3}$) by the helium partial pressure within the ampoule, which can be related to the atomic abundance of helium in the gas phase using the ideal gas law and the formula for the volume of a cylinder:

$$C_{He}^{glass} = S(T) \bullet P_{He}^{gas}$$

$$P_{He}^{gas} = \frac{n_{gas} RT}{V_{gas}}$$
$$V_{gas} = \pi a^2 h$$

$$C_{He}^{glass} = S(T) \bullet \frac{n_{gas} RT}{\pi a^2 h} \tag{13}$$

For helium permeation out of glass ampoules, the internal pressure is not constant. n_{gas} changes after every step and Eq. (13) is reapplied to the inner boundary cell where r = a. While the helium concentrations in the glass and in the encapsulated gas phase are both dependent on the length of the ampoule (h), they compensate for one another such that the absolute atom flux into and out of the ampoule is not a function of h.

In order to simulate permeation, the atomic abundances of helium within the gas phases both outside the ampoule (at high vacuum) and inside the ampoule must be tracked, as do the fluxes between the glass and these two gas phases. These fluxes were calculated simply as the diffusive flux in $\frac{atoms}{s \bullet cm^2}$ as is given by Fick's law. For spatially discrete cells throughout the ampoule at any radius except for the boundaries, the diffusive flux can be approximated with the finite difference:

$$J_r = -D\frac{\partial C_r}{\partial r} \approx \ -D\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r} \tag{14}$$

However, Eq. (14) cannot be applied at the ampoule's inner wall when r = a or outer wall when r = b because there is no remaining glass on one side of the cell. In that case, second order terms from Taylor expansions centered at $a + \Delta r$ and $b - \Delta r$ can be used to better approximate the $\frac{\partial C_a}{\partial r}$ and $\frac{\partial C_b}{\partial r}$ terms. Letting $f(r) = \ \frac{\partial C_r}{\partial r}$ and $f'(r) = \ \frac{\partial^2 C_r}{\partial r^2}$:

$$J_a = -D\frac{\partial C_a}{\partial r} \approx \ -D\left(\frac{\partial C_{a+\Delta r}}{\partial r} + \frac{\partial^2 C_{a+\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\ \frac{C_{a+2\Delta r} - C_a}{2\Delta r} + \frac{C_a - 2C_{a+\Delta r} + C_{a+2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{15}$$

$$J_b = -D\frac{\partial C_b}{\partial r} \approx \ -D\left(\frac{\partial C_{b-\Delta r}}{\partial r} - \frac{\partial^2 C_{b-\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_b - C_{b-2\Delta r}}{2\Delta r} - \frac{C_b - 2C_{b-\Delta r} + C_{b-2\Delta r}}{2\Delta r^2}\Delta r\ \right) \tag{16}$$

The helium contents in the measurable (external, high vacuum) reservoir and the inner encapsulated gas phase are updated every step using fluxes determined from Eq. (16) and (15) respectively. To convert these area-normalized fluxes into absolute atomic abundances, these fluxes are multiplied by the time increment (Δt) and the surface area of the cylinder at the outer and inner walls respectively.

Simulations require an initial helium concentration profile in the glass appropriate for a fresh charge of helium at a given temperature. A closed-form analytical solution for this profile is relatively easy to derive, but such a derivation is of little significance to this work and is too lengthy to include here. Therefore, the steady state concentration profile is retrieved numerically: the inner and outer boundary conditions are held fixed while the model iterates Eq. (12) until the concentration profile no longer changes. This produces the same result as solving Eq. (11) analytically for the steady-state profile.

4. Goethite dehydroxylation: thermodynamic model

The following sections give the full formulation, data and sensitivity tests behind the condensed analysis of goethite dehydroxylation in the main text (§2.6 and §4.6).

Goethite breaks down to hematite by the dehydroxylation reaction

$$2\,\alpha\text{-FeOOH} \;\rightleftharpoons\; \alpha\text{-Fe}_2\text{O}_3 + \text{H}_2\text{O}_{(\text{fluid})}, \tag{17}$$

whose Gibbs energy change, written per mole of hematite formed, is

$$\Delta G_r(T,P) = G_{\text{hem}}(T,P) + G_{\text{w}}(T,P) - 2\,G_{\text{goe}}(T,P) + \Delta G_{\text{surf}}. \tag{18}$$

Goethite is stable where $\Delta G_r > 0$ and hematite plus water where $\Delta G_r < 0$. We use the apparent Gibbs energy convention, in which element reference terms cancel in the balanced reaction. For each solid,

$$G_i(T,P) = \Delta_f H_i^0 + \int_{T_r}^{T} C_{p,i}\,dT \;-\; T\left[S_i^0 + \int_{T_r}^{T} \frac{C_{p,i}}{T}\,dT\right] \;+\; V_i\,(P - P_r), \tag{19}$$

with $T_r = 298.15$ K and $P_r = 1$ bar, treating the solids as incompressible with no thermal expansion (the $V\,\Delta P$ term never exceeds a few kJ below

1 GPa). Goethite heat capacity uses the Berman and Brown (1985) form refined by Majzlan et al. (2003a),

$$C_p = k_0 + k_1 T^{-1/2} + k_2 T^{-2} + k_3 T^{-3}, \tag{20}$$

which integrates to

$$\begin{aligned}\int_{T_r}^{T} C_p\,dT &= k_0(T - T_r) + 2k_1\left(T^{1/2} - T_r^{1/2}\right) \\ &\quad - k_2\left(T^{-1} - T_r^{-1}\right) - \tfrac{1}{2}k_3\left(T^{-2} - T_r^{-2}\right), \qquad (21)\\ \int_{T_r}^{T} \frac{C_p}{T}\,dT &= k_0 \ln\frac{T}{T_r} - 2k_1\left(T^{-1/2} - T_r^{-1/2}\right) \\ &\quad - \tfrac{1}{2}k_2\left(T^{-2} - T_r^{-2}\right) - \tfrac{1}{3}k_3\left(T^{-3} - T_r^{-3}\right). \qquad (22)\end{aligned}$$

Hematite uses a Maier–Kelley form, $C_p = a + bT + cT^{-2}$, with

$$\int_{T_r}^{T} C_p\,dT = a(T - T_r) + \tfrac{1}{2}b\left(T^2 - T_r^2\right) - c\left(T^{-1} - T_r^{-1}\right), \tag{23}$$

$$\int_{T_r}^{T} \frac{C_p}{T}\,dT = a\ln\frac{T}{T_r} + b(T - T_r) - \tfrac{1}{2}c\left(T^{-2} - T_r^{-2}\right). \tag{24}$$

Water is evaluated with the IAPWS-95 equation of state (Wagner and Pruß, 2002), implemented in CoolProp (Bell et al., 2014), which is valid for liquid, vapor and supercritical water to 1 GPa:

$$G_{\mathrm{w}}(T,P) = \Delta_f H^0_{\mathrm{w}} + \left[h(T,P) - h(T_r,P_r)\right] - T\left[S^0_{\mathrm{w}} + s(T,P) - s(T_r,P_r)\right], \tag{25}$$

where h and s are the IAPWS-95 molar enthalpy and entropy and the reference state is liquid water at T_r, P_r. Table S1 lists all input data. We use the enthalpies and entropies of Majzlan et al. (2003a) refined by mathematical programming against the phase-equilibrium brackets of Voigt and Will (1981).

As a check, at 298.15 K and 1 bar with liquid water, Eqs. (18)–(19) give $\Delta G_r = -0.26$ kJ, i.e. hematite plus liquid water is favored by only 0.13 kJ per mole of FeOOH. This is well inside the ± 1.4 kJ mol^{-1} uncertainty that Majzlan et al. (2003a) assign to the goethite–hematite free-energy difference; the two assemblages are essentially balanced at ambient conditions. We carry this uncertainty (± 2.8 kJ per reaction as written) as bounding curves on the phase diagram.

Table S1: Thermodynamic data used for Eq. (18).

	Goethite (α-FeOOH)	Hematite (α-Fe_2O_3)
$\Delta_f H^0$ (kJ mol^{-1})	-561.4	-825.8
S^0 (J mol^{-1} K^{-1})	59.5	87.4
V^0 (cm^3 mol^{-1})	20.881	30.274
C_p coefficients	$k_0 = 133.8765$	$a = 98.28$
	$k_1 = -738.5577$	$b = 7.782 \times 10^{-2}$
	$k_2 = -2.177657 \times 10^6$	$c = -1.485 \times 10^6$
	$k_3 = 2.165124 \times 10^8$	
C_p(298 K) (J mol^{-1} K^{-1})	74.8	104.8
$\gamma_{\mathrm{hydrated}}$ (J m^{-2})	0.60 ± 0.10	0.75 ± 0.16
$\gamma_{\mathrm{anhydrous}}$ (J m^{-2})	0.91 ± 0.09	1.90 ± 0.30

Enthalpy, entropy and C_p: Majzlan et al. (2003b,a); surface energies: Navrotsky et al. (2008). Water: $\Delta_f H^0_{\mathrm{w}} = -285.83$ kJ mol^{-1}, $S^0_{\mathrm{w}} = 69.95$ J mol^{-1} K^{-1}; IAPWS-95 for $h(T,P)$, $s(T,P)$.

Following Navrotsky et al. (2008), a finite grain adds a surface term $G = G_{\mathrm{bulk}} + \gamma A$, where γ is the surface energy (J m^{-2}) and A the molar surface area. For the reaction,

$$\Delta G_{\mathrm{surf}} = \gamma_{\mathrm{hem}} A_{\mathrm{hem}} - 2\,\gamma_{\mathrm{goe}} A_{\mathrm{goe}}. \tag{26}$$

We compare the phases at equal surface area per mole of Fe, the basis of the Navrotsky et al. (2008) stability diagrams. Physically, this corresponds to pseudomorphic dehydroxylation, in which hematite inherits the external envelope of the parent goethite. For spherical goethite grains of diameter d,

$$A_{\mathrm{Fe}} = \frac{6\,V_{\mathrm{goe}}}{d}, \qquad \Delta G_{\mathrm{surf}} = 2\,A_{\mathrm{Fe}}\,(\gamma_{\mathrm{hem}} - \gamma_{\mathrm{goe}})\,. \tag{27}$$

The result depends on the choice of basis: at equal diameter, the same hydrated surface energies destabilize fine goethite (§4.2). We use hydrated surface energies throughout. They are referenced to liquid water, so they apply wherever the surfaces are in equilibrium with water at or near saturation, which includes all of the hydrothermal experiments. Anhydrous surface energies are relevant only to surfaces that have lost their chemisorbed water, i.e. under vacuum.

For each temperature we solve $\Delta G_r(T,P) = 0$ for $P = P_{\mathrm{H_2O}}$ numerically (Brent's method), with the solids at the fluid pressure. Where water behaves as an ideal vapor, the boundary reduces to

$$\log_{10} P_{\mathrm{eq}}\ (\mathrm{bar}) \approx -\frac{\Delta G_r^{0,\mathrm{vap}}(T) + \Delta G_{\mathrm{surf}}}{\ln(10)\,RT}, \tag{28}$$

where $\Delta G_r^{0,\mathrm{vap}}$ refers to water vapor at 1 bar. The quantity that matters for kinetics is the driving force for hematite formation, $\Delta\mu \equiv -\Delta G_r$ (per mole of Fe_2O_3). Water enters Eq. (18) as a product, so $\Delta\mu$ is controlled by the water chemical potential. In the ideal-vapor limit, lowering P_{H_2O} from saturation to P raises the driving force by

$$\Delta\mu(P) - \Delta\mu(P_{\mathrm{sat}}) \approx RT \ln \frac{P_{\mathrm{sat}}}{P}. \tag{29}$$

All reported driving forces use the full IAPWS-95 calculation rather than this approximation.

4.1. Stability and driving force at the experimental conditions

The thermodynamic model above places the goethite–hematite–water boundary in $\log P_{H_2O}$–T space (Fig. S2). For $d = 10$ nm, $A_{\mathrm{Fe}} = 6 \times 20.881 \times 10^{-6}/10^{-8} = 1.253 \times 10^4$ m^2 mol^{-1}, and with hydrated surface energies $\Delta G_{\mathrm{surf}} = 2 \times 1.253 \times 10^4 \times (0.75 - 0.60) = +3.76$ kJ (Eq. (27)). The corresponding values are +0.38 kJ at 100 nm and +0.04 kJ at 1 µm. A positive ΔG_{surf} stabilizes goethite. Above the saturation pressure, water is liquid, the reaction volume $\Delta V_r = V_{\mathrm{hem}} + V_{\mathrm{w}} - 2V_{\mathrm{goe}} \approx +6.6$ cm^3 is small, and the boundary becomes nearly vertical in $\log P$–T space. Bulk goethite meets the liquid–vapor curve at 18 °C, 100 nm goethite at 28 °C and 10 nm goethite at 108 °C. At 25 °C, bulk goethite requires $P_{H_2O} \approx 10^{2.6}$ bar (compressed liquid). By contrast, 100 nm goethite is stable above $10^{-1.52}$ bar (96% relative humidity) and 10 nm goethite above $10^{-2.11}$ bar (24% relative humidity). Bulk goethite is therefore metastable at the conditions of every experiment considered here, in water or in vacuum.

Stability alone does not determine what happens in a step heat; the rate of conversion does, and that depends on the driving force $\Delta\mu$. At 250 °C, for example, Eq. (29) gives $RT \ln(39.76 \text{ bar}/1.013 \times 10^{-8} \text{ bar}) = 4.35 \times 22.1 = 96.1$ kJ between water saturation and the vacuum of the step-heating line. The full IAPWS-95 calculation gives $106.0 - 10.65 = 95.4$ kJ; the difference reflects the non-ideality of saturated steam. Table S2 gives bulk driving forces at the experimental temperatures. At saturation, $\Delta\mu$ is 6–11 kJ, only 1.6–2.4 times RT. Under the vacuum of a step-heating line ($P_{H_2O} \approx 10^{-8}$ atm) it is 76–106 kJ, about ten times larger. The chemical potential of water is therefore the key variable separating the two experimental settings.

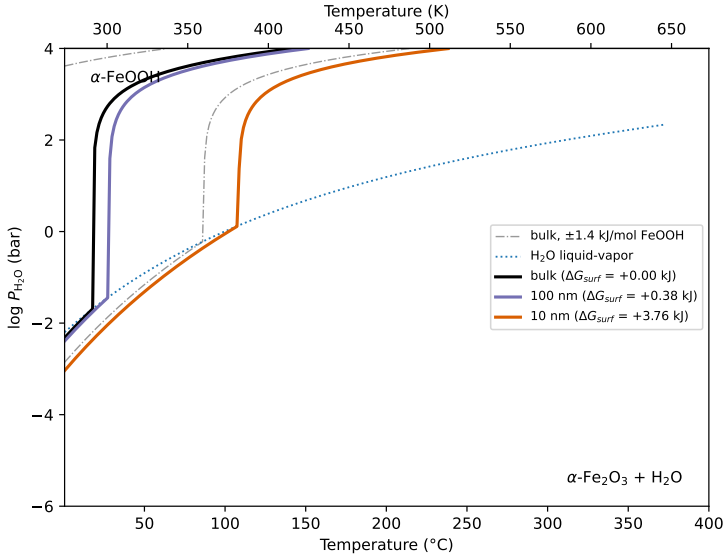


Figure S2: Goethite–hematite–water equilibrium in $\log P_{\mathrm{H_2O}}$–T space for bulk goethite and for 100 and 10 nm grains with hydrated surfaces, compared at equal surface area per mole of Fe (Eq. (27)). Goethite is stable above and to the left of each curve. Dash-dot curves: bulk boundary shifted by the ± 1.4 kJ mol^{-1} FeOOH uncertainty of Majzlan et al. (2003a). Dotted: liquid–vapor curve of water. Thermodynamic data in Table S1.

Table S2: Bulk driving force for hematite formation, $\Delta\mu = -\Delta G_r$ (kJ per mole Fe_2O_3), at water saturation and at $P_{\mathrm{H_2O}} = 10^{-8}$ atm.

T (°C)	P_{sat} (bar)	$\Delta\mu_{\mathrm{sat}}$	$\Delta\mu_{\mathrm{sat}}/RT$	$\Delta\mu_{\mathrm{vac}}$	$(\Delta\mu_{\mathrm{vac}}/\Delta\mu_{\mathrm{sat}})^2$
150	4.76	5.68	1.61	75.8	178
175	8.93	6.88	1.84	83.4	147
200	15.55	8.10	2.06	91.0	126
225	25.50	9.36	2.26	98.5	111
250	39.76	10.65	2.45	106.0	99

4.2. Sensitivity of the size effect to the surface-area basis

The sign of the surface stabilization depends on how the surface areas of the two phases are related. At equal area per mole of Fe (Eq. (27)), goethite is stabilized whenever $\gamma_{\mathrm{hem}} > \gamma_{\mathrm{goe}}$. If both phases are instead spheres of the same diameter d, then $A_i = 6V_i/d$ and

$$\Delta G_{\mathrm{surf}} = \frac{6}{d}\left(\gamma_{\mathrm{hem}} V_{\mathrm{hem}} - 2\gamma_{\mathrm{goe}} V_{\mathrm{goe}}\right), \tag{30}$$

which stabilizes goethite only if

$$\frac{\gamma_{\mathrm{hem}}}{\gamma_{\mathrm{goe}}} > \frac{2V_{\mathrm{goe}}}{V_{\mathrm{hem}}} = \frac{2 \times 20.881}{30.274} = 1.379. \tag{31}$$

Hematite is denser and carries less surface per Fe at equal size. For hydrated surfaces, $\gamma_{\mathrm{hem}}/\gamma_{\mathrm{goe}} = 0.75/0.60 = 1.25 < 1.379$, so fine goethite is destabilized: at 10 nm, $\Delta G_{\mathrm{surf}} = (6/10^{-8})(0.75 \times 30.274 - 2 \times 0.60 \times 20.881) \times 10^{-6} = -1.41$ kJ. For anhydrous surfaces, $1.90/0.91 = 2.09 > 1.379$ and $\Delta G_{\mathrm{surf}} = +11.7$ kJ at 10 nm. Table S3 compares the two bases. The equal-area basis corresponds to pseudomorphic conversion and underlies the main-text calculations. The equal-diameter basis is more appropriate where hematite grows as separate crystals, as in fluid-mediated conversion (§8). There the surface term slightly favors hematite, but it is negligible at micrometer sizes.

Table S3: ΔG_{surf} (kJ per reaction as written) for the two surface-area bases.

	Hydrated		Anhydrous	
d	equal area	equal diameter	equal area	equal diameter
1 μm	+0.04	−0.01	+0.25	+0.12
100 nm	+0.38	−0.14	+2.48	+1.17
10 nm	+3.76	−1.41	+24.8	+11.7

5. Kinetic analysis of the vacuum step heats

Two published vacuum step heats of irradiated Win-06-01b are reanalyzed here. The experiment of Farley et al. (2024) (hereafter KAF) used 1 h steps from 70 to 330 °C, including isothermal repeats at 210, 230 and 240 °C. Its extraction line was open to a liquid-nitrogen-cooled charcoal trap, so evolved water did not accumulate, and the water released in each step was measured

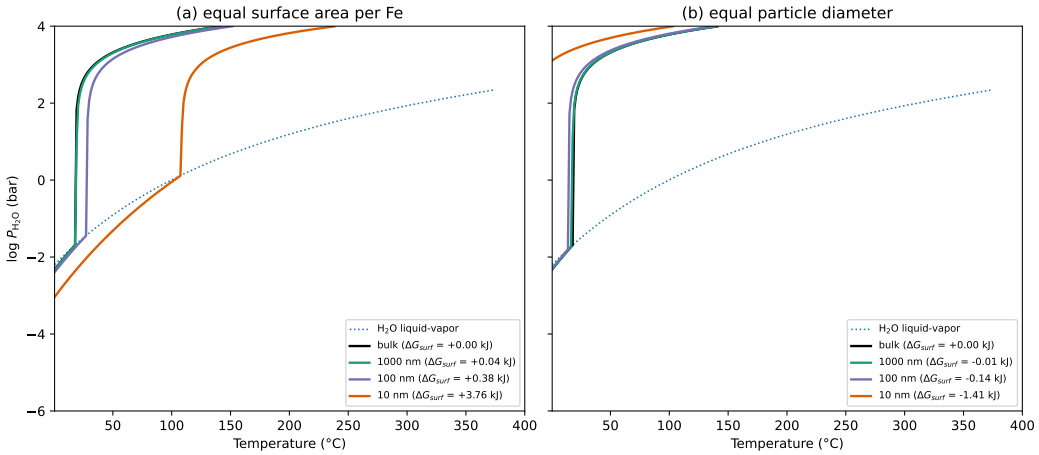


Figure S3: Goethite–hematite–water boundaries for hydrated surfaces at (a) equal surface area per mole of Fe and (b) equal particle diameter. The size effect changes sign between the two bases.

by capacitance manometry. The experiment of Monteiro (2017) (hereafter HDM) used a different schedule: prograde to 238 °C, retrograde to 180 °C, then prograde again, with 0.5 h steps above 238 °C, to 498 °C. Water could accumulate in the HDM line. For each experiment, the cumulative fraction of ^{3}He released after step i is

$$F_i = \frac{\sum_{j \le i} {}^3\mathrm{He}_j}{\sum_{\mathrm{steps}} {}^3\mathrm{He} + {}^3\mathrm{He}_{\mathrm{fusion}}}. \tag{32}$$

We describe conversion by a first-order rate law,

$$\frac{d\alpha}{dt} = k(T)\,(1-\alpha), \qquad k(T) = A \exp\!\left(-\frac{E_a}{RT}\right), \tag{33}$$

where α is the fraction of goethite converted. For a powder of many grains, first-order kinetics is the limit in which each grain converts rapidly once a hematite nucleus forms, so that the conversion rate is set by the nucleation rate per untransformed grain (Mampel, 1940). k is then a per-grain nucleation rate. We assume ^{3}He is released when its host grain converts, except for a fraction ϕ that is inherited by the hematite product:

$$F = (1-\phi)\,\alpha. \tag{34}$$

For a schedule of isothermal steps of duration Δt_j at T_j, Eq. (33) integrates to

$$\alpha_i = 1 - \exp\left[-A \sum_{j \le i} m_j \, \Delta t_j \exp\left(-\frac{E_a}{R T_j}\right)\right], \tag{35}$$

where $m_j = s$ for the two KAF steps run with the trap isolated (§5) and $m_j = 1$ otherwise. Inverting Eq. (33) step by step gives an apparent rate constant for every step that is independent of any fitted parameter except ϕ:

$$k_i = -\frac{1}{\Delta t_i} \ln \frac{1 - \alpha_i}{1 - \alpha_{i-1}}, \qquad \alpha_i = \frac{F_i}{1 - \phi}. \tag{36}$$

For comparison, the diffusion parameter of each step is obtained with the same Fechtig and Kalbitzer (1966) inversion for a sphere that is applied to the hydrothermal experiment (§2.1),

$$\frac{Dt}{a^2} = \begin{cases} \dfrac{1}{\pi^2}\left[2\pi - \dfrac{\pi^2}{3}F - 2\pi\sqrt{1 - \dfrac{\pi F}{3}}\right], & F \le 0.85, \\ -\dfrac{1}{\pi^2}\ln\left[\dfrac{\pi^2}{6}(1 - F)\right], & F > 0.85, \end{cases} \tag{37}$$

and

$$\left(\frac{D}{a^2}\right)_i = \frac{(Dt/a^2)_i - (Dt/a^2)_{i-1}}{\Delta t_i}.$$

For the hydrothermal experiment, F_i and D/a^2 are always taken from the bounding-method reduction (main text §2.7.5): the cumulative release at the end of step i is bounded by the measured cumulative ^{3}He and by that value plus $n_{t,\mathrm{UB}}$, and the central value is their midpoint. As in the main text, steps without defined bounds and steps with less than 0.1 pccSTP of measured ^{3}He are excluded.

All steps are plotted, but fits use only KAF steps 5–23 (185–285 °C) and HDM steps 8–28 (199–338 °C). KAF steps 1–4 and HDM steps 2–7 (below ∼180 °C) release a small ($< 2\%$), separate low-temperature component (§5). Later steps sit on the post-conversion plateau, where $1 - \alpha \to 0$ and Eq. (36) becomes controlled by the choice of ϕ. The KAF steps run with the trap isolated are excluded from the Arrhenius regression.

5.1. Results

Fitting Eqs. (34)–(35) to the cumulative KAF ^{3}He release by least squares gives $E_a = 122.9$ kJ mol^{-1}, $\ln A = 18.67$ (A in s^{-1}), $\phi = 0.112$ and $s = 0.46$, with an rms misfit in F of 0.0085. The step rate constants of both experiments (Eq. (36)) fall on a single Arrhenius line (main text Fig. 10a). This includes the HDM retrograde steps, which a first-order process requires: its rate depends only on T and on the amount of goethite remaining. An unweighted least-squares fit of $\ln k_i$ against $1/T_i$ for the 38 steps in the fitting window gives

$$E_a = 121.4 \pm 4.4 \text{ kJ mol}^{-1}, \qquad \ln A = 18.08 \pm 1.05 \tag{38}$$

(KAF alone 134.9 ± 6.1; HDM alone 116.3 ± 5.6 kJ mol^{-1}), in agreement with the cumulative fit. At 250 °C this gives $k = 5.4 \times 10^{-5}$ s^{-1}, or 18% conversion per hour.

In the diffusion representation (main text Fig. 10b) the two experiments do not collapse. HDM's retrograde steps between 180 and 220 °C plot about three natural-log units above HDM's own prograde steps at the same temperatures. Volume diffusion from grains that are being progressively depleted cannot produce this, whether from a single domain or multiple domains: retrograde steps must fall on or below the prograde trend. D/a^2 also turns over above ∼275 °C, where release stalls at the plateau.

The parameters fitted to KAF alone predict the independent HDM schedule with an rms misfit in F of 0.017 over the fitting window (Fig. S4c,d). The prediction reproduces the main release and follows the retrograde–prograde cycle within about a factor of two on individual steps. A single-domain spherical diffusion model fits KAF worse (rms 0.046) and predicts HDM worse (rms 0.053). It also predicts $F \to 1$ by ∼300 °C, whereas both experiments stall at $F \approx 0.89$. Within a single schedule, fits alone cannot discriminate between kinetic mechanisms (§5.2). The second schedule is what gives the comparison its power.

During the main vacuum release (KAF, $0.04 < F < 0.88$), the step ^{4}He/^{3}He ratio stays within 0.95–1.10 of the bulk ratio. Grains therefore release their integrated inventory as they convert, rather than releasing a diffusive, rim-weighted component.

In both experiments release stalls at $F \approx 0.89$. The fitted $\phi = 0.112$ is the same for KAF and HDM. In HDM, the remaining ∼11% is released slowly between 340 and 500 °C and on fusion. We interpret this fraction as ^{3}He

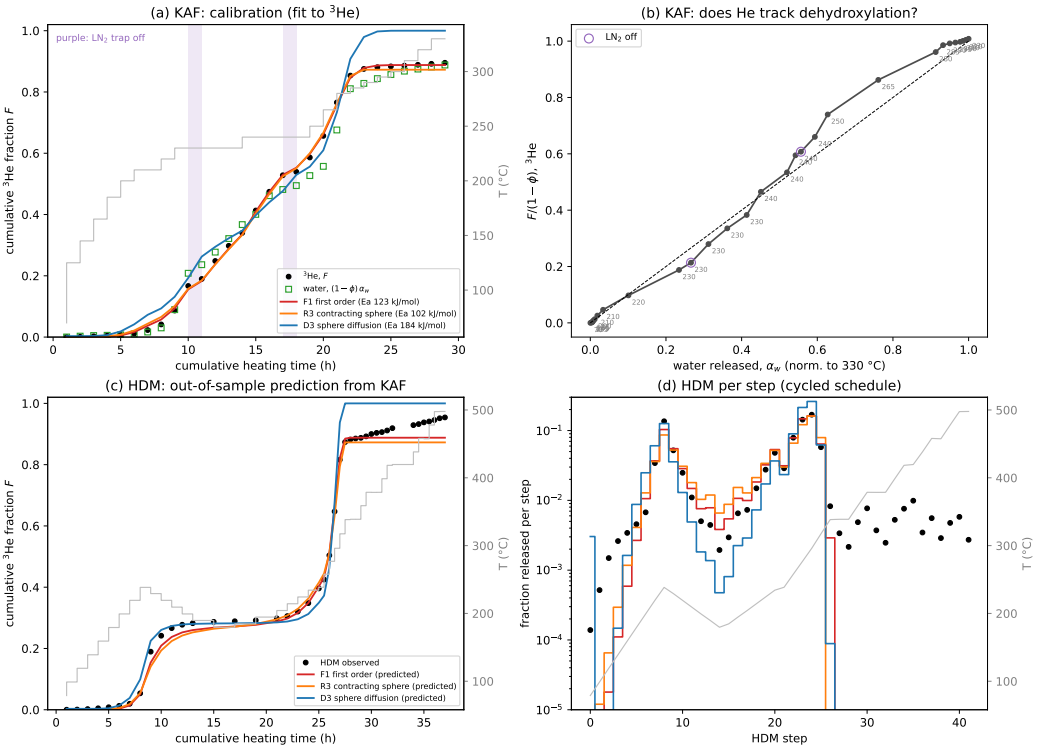


Figure S4: (a) Cumulative ^{3}He release of KAF (Farley et al., 2024) with fits of the first-order (F1), contracting-sphere (R3) and spherical-diffusion (D3) models; purple bands mark the steps with the LN_2 trap isolated; grey: temperature schedule. (b) ^{3}He against water released during KAF, both normalized. (c) HDM (Monteiro, 2017) cumulative release and the predictions from the KAF-calibrated models, with no parameters refitted. (d) HDM per-step release fractions.

inherited by the hematite product and subsequently released by the much slower diffusion out of hematite. Single-domain goethite diffusion cannot produce a plateau. The conversion and the plateau correspond to the "Stage 2" and "Stage 3" release of Farley et al. (2024), whose transition near $F \approx 0.88$ is independent of heating schedule.

The water evolved in each KAF step tracks the ^{3}He release closely (Fig. S4b). Water leads ^{3}He slightly at 230 °C and lags it by ∼0.1 between 250 and 280 °C. Fitting Eq. (35) to the water data alone gives $E_a = 91.5$ kJ mol^{-1} (rms 0.033). The larger misfit is consistent with water adsorbed before dehydroxylation (∼1% of the total, released below 200 °C). It is also consistent with systematically low manometer readings on the first step of each measurement day, and with normalizing the water to the 330 °C total. Water release thus provides an independent measure of conversion that is consistent with the ^{3}He record.

In two KAF steps the trap was isolated and evolved water accumulated in the line. At 230 °C the step rate constant fell to 0.33 times the mean of its neighbors at the same temperature, and at 240 °C to 0.22 times. The rate recovered fully when the trap was restored. Water and ^{3}He were suppressed together. A diffusion process would not respond to the partial pressure of water. We use these two steps in §7 to calibrate the nucleation model.

Below ∼180 °C both vacuum experiments release ^{3}He 0.5–3 orders of magnitude faster than the extrapolated conversion rate (Fig. S5). This component amounts to less than 2% of the ^{3}He and is not conversion. We attribute it to diffusion from the least retentive domains (main text §4.5).

5.2. Kinetic mechanisms and the limits of a single heating schedule

Solid-state rate laws take the form $d\alpha/dt = A\exp(-E_a/RT)\,h(\alpha)$. For any heating schedule this integrates to

$$g(\alpha) \equiv \int_0^\alpha \frac{d\alpha'}{h(\alpha')} = x, \qquad x = A\sum_j \Delta t_j \exp\left(-\frac{E_a}{RT_j}\right). \tag{39}$$

We tested the mechanisms in Table S4, including spherical diffusion on the same footing. Under vacuum, the driving-force factor $1 - \exp(-\Delta\mu/RT)$ is unity to better than 10^{-9}, so x is simply an Arrhenius-weighted time. Diffusion and conversion then differ only in the shape $\alpha(x)$ and in the fitted E_a.

Table S4: Kinetic models and their integrated forms.

Model	$h(\alpha)$	$g(\alpha)$	$\alpha(x)$
R2 contracting cylinder	$2(1-\alpha)^{1/2}$	$1-(1-\alpha)^{1/2}$	$1-(1-x)^2$
R3 contracting sphere	$3(1-\alpha)^{2/3}$	$1-(1-\alpha)^{1/3}$	$1-(1-x)^3$
F1 first order	$1-\alpha$	$-\ln(1-\alpha)$	$1-e^{-x}$
A2, A3 Avrami ($n=2,3$)	$n(1-\alpha)[-\ln(1-\alpha)]^{1-1/n}$	$[-\ln(1-\alpha)]^{1/n}$	$1-e^{-x^n}$
D3 sphere diffusion	—	—	$1-\frac{6}{\pi^2}\sum_n \frac{e^{-n^2\pi^2 x}}{n^2}$

Synthetic test.. We generated noiseless release curves from the R3 model ($E_a = 150$ kJ mol^{-1}) on two schedules: a monotonic series of 1 h steps from 150 to 390 °C, and a cycled schedule with retrograde steps and long isothermal holds. We then fitted every model with E_a and A free. Table S5 shows that every model except the Avrami forms fits within 1% in F by adjusting E_a. With realistic analytical uncertainty, goodness of fit to a single schedule cannot identify the mechanism. The fitted activation energy trades off against the shape of $\alpha(x)$. Discrimination requires a second, independent schedule, an independent measure of α, or both. The vacuum experiments supply both (§5).

Table S5: Fits to synthetic R3 data (E_a in kJ mol^{-1}; rms misfit in F).

	Monotonic schedule		Cycled schedule	
Model	E_a	rms	E_a	rms
R2	140.0	0.0031	140.4	0.0028
R3 (true)	150.0	0.0000	150.0	0.0000
F1	173.0	0.0074	177.4	0.0092
A2	81.7	0.0073	83.8	0.0077
A3	51.3	0.0073	53.8	0.0062
D3	296.2	0.0026	296.9	0.0025

Fits to the vacuum data.. Table S6 gives the fit of each model to the cumulative KAF ^{3}He release within the fitting window. It also gives the misfit of the resulting out-of-sample prediction of HDM, and the fit to the KAF water data. F1 gives the best fit and the best prediction. R3 and R2 are close behind, so we describe the mechanism as nucleation- or interface-controlled rather than claim first order specifically. The Avrami fits reach the lower

bound of E_a and fail on both data sets. Spherical diffusion fits KAF five times worse than F1 and cannot produce the $F \approx 0.89$ plateau.

Table S6: Model fits to the vacuum experiments (from `vacuum_kinetics_KAF_to_HM_fits.csv`). s: rate factor for the trap-isolated steps.

Model	E_a	$\ln A$	ϕ	s	rms KAF	rms HDM	E_a (water)
R2	90.3	10.00	0.132	0.54	0.0133	0.0292	64.7
R3	101.7	12.38	0.127	0.50	0.0111	0.0234	73.4
F1	122.9	18.67	0.112	0.46	0.0085	0.0167	91.5
A2	25.6	−4.70	0.000	0.20	0.0207	0.1814	20.2
A3	20.0[a]	−5.89	0.159	0.00	0.0446	0.2348	20.0[a]
D3	183.7	29.50	—	1.70	0.0455	0.0529	180.2

E_a in kJ mol^{-1}, A in s^{-1}. [a]At the lower bound of the fit.

6. Hydrothermal tests of diffusion versus conversion

The hydrothermal experiment heated an aliquot of the same goethite in distilled water, with much of the sample submerged in the liquid. The schedule rose from 170 to 230 °C, cycled down to 175 °C, rose to 250 °C, cycled down to 225 °C and back to 250 °C, where three 5 h steps preceded a 170.8 h step, and ended with a step at 240 °C (442.8 h in total). Only 14.7% of the ^{3}He was released. Under the vacuum kinetics of Eq. (38), the same schedule would have converted the sample completely. Two tests, both applied to the bounding-method reduction (main text §2.7.5), show that the release follows diffusion rather than conversion.

Test 1: history dependence through temperature cycles. A line through the first prograde sweep (200–230 °C) serves as the reference. The two retrograde steps of the first cycle with adequate signal, at 225 and 215 °C, fall 0.45 and 0.86 natural-log units below it in $\ln(D/a^2)$ (Eq. (37); central values). The upper bound of each step still falls 0.18 and 0.47 log units below a line fitted to the lower bounds of the first sweep. On the next prograde sweep, the 220 °C step sits 0.58 log units (at least 0.32) below its first-sweep value (main text Fig. 11a). The low-signal steps of the first retrograde sweep (205–175 °C, and 200–210 °C on the following prograde sweep; excluded from all plots) sit lower still, by 1.3–2.1 log units. In the last cycle, retrograde and prograde values at 230–245 °C agree within 0.07–0.45 log units, and their bounds overlap at every temperature. The Stage 1 regression over this interval (cumulative

yield ≥7%) gives $E_a = 157.1 \pm 5.3$ kJ mol^{-1} (§2.1). This pattern is the signature of depleting fast, low-retentivity domains in the first cycle and diffusion dominated by more retentive domains thereafter. A conversion rate controlled at grain surfaces or interfaces has almost no memory at $F \approx 0.04$, because almost no surface has been consumed. It would return retrograde steps to the prograde line, as the vacuum HDM experiment does in the reaction representation.

Test 2: the 250 °C holds. Diffusion slows as the diffusion front moves inward; conversion controlled at an interface proceeds at a nearly constant rate while $\alpha \ll 1$. We calibrate both on the three 5 h steps at 250 °C and predict the 170.8 h step. For diffusion, the mean of the three steps' $\ln(D/a^2)$ is −19.91 (bounds −20.13 to −19.71), i.e. $D/a^2 = 2.26 \times 10^{-9}$ s^{-1}. The Fourier number at the start of the long step, $(Dt/a^2)_0$, follows from Eq. (37) with the central $F_0 = 0.0874$; then

$$\left(\frac{Dt}{a^2}\right)_{\text{end}} = \left(\frac{Dt}{a^2}\right)_0 + \frac{D}{a^2}\,\Delta t, \qquad F_{\text{end}} = 1 - \frac{6}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\exp\left[-n^2\pi^2\left(\frac{Dt}{a^2}\right)_{\text{end}}\right]. \tag{40}$$

For an interface-controlled reaction (contracting sphere, $d\alpha/dt \propto (1-\alpha)^{2/3}$), the release rate over the long step is the mean rate of the three 5 h steps, scaled by $(1-\alpha)^{2/3}$ from the calibration window to the long step (a ∼3% correction). The bounding method limits the ^{3}He released by the sample during the three 5 h steps to 1.08–1.22 pccSTP, and during the long step to 8.52–8.65 pccSTP. Table S7 compares the predictions. Diffusion predicts the long step within its bounds, and the observed release lies inside the diffusion envelope. The constant-rate reaction overpredicts the release in the long step by 39–60% (12.0–13.6 pccSTP).

Table S7: The 170.8 h step at 250 °C, predicted from the three preceding 5 h steps using the bounding-method reduction. Ranges span the lower and upper bounds.

	F at end of step
Start of step	0.0872–0.0877
Observed	0.1460–0.1463
Diffusion (sphere, D/a^2 from 5 h steps)	0.148 (0.139–0.158)
Interface-controlled reaction (constant rate)	0.170–0.180

Plotted as first-order rate constants (main text Fig. 10a), the hydrothermal steps sit 1.3–2.8 orders of magnitude (median 2.4) below the vacuum

line. They show what a first-order process forbids: retrograde steps 0.7–1.5 natural-log units below prograde steps at the same temperature, and a falling k during the 250 °C holds. In the diffusion representation (main text Fig. 10b) the same steps form a coherent, history-dependent array. The vacuum experiments behave like conversion and the hydrothermal experiment behaves like diffusion. The conversion rate expected at water saturation is too small, by tens of orders of magnitude, to contribute to the observed release (§7).

7. Nucleation model and calibration of the effective interfacial energy

In classical nucleation theory, a spherical hematite nucleus of radius r has free energy

$$\Delta G(r) = -\frac{4}{3}\pi r^3 \frac{\Delta\mu}{V_{\mathrm{hem}}} + 4\pi r^2 \gamma, \tag{41}$$

where $\Delta\mu$ is the driving force per mole of Fe_2O_3 (Table S2), V_{hem} the molar volume and γ the energy of the interface surrounding the nucleus. Setting $d\Delta G/dr = 0$ gives the barrier height

$$\Delta G^* = \frac{16\pi\,\gamma^3\,V_{\mathrm{hem}}^2}{3\,\Delta\mu^2}. \tag{42}$$

Written per formula unit, $V_{\mathrm{hem}} \to V_{\mathrm{hem}}/N_A$ and $\Delta\mu \to \Delta\mu/N_A$, so Avogadro's number cancels and Eq. (42) is directly the barrier per nucleus in joules. In the first-order limit of §5, the rate constant is a nucleation rate,

$$k = A\exp\!\left(-\frac{E_a}{RT}\right)\exp\!\left(-\frac{\Delta G^*}{k_B T}\right),$$

$$\frac{\Delta G^*}{k_B T} = \frac{C\gamma^3}{T\,\Delta\mu^2}, \qquad C \equiv \frac{16\pi V_{\mathrm{hem}}^2}{3k_B} = 1.112\times10^{15}\ \mathrm{m^6\,K\,J^{-1}\,mol^{-2}}. \tag{43}$$

The barrier depends on $\gamma^3/\Delta\mu^2$. Between vacuum and water saturation, $\Delta\mu$ falls tenfold and the barrier rises by the factor in the last column of Table S2, about 100 at 250 °C and nearly 180 at 150 °C.

Nucleation must be internal. The surface energies of Navrotsky et al. (2008) show where hematite cannot nucleate. With a free hematite surface as the interface, at 250 °C, Eq. (43) gives $\Delta G^*/k_B T \approx 1.3\times10^3$ under vacuum ($\gamma = 1.90\ \mathrm{J\,m^{-2}}$, $\Delta\mu = 106.0$ kJ). At saturation it gives $\approx 7.9\times10^3$

($\gamma = 0.75$ J m^{-2}, $\Delta\mu = 10.65$ kJ). Nucleation on a laboratory timescale requires roughly $\Delta G^*/k_BT \lesssim 20$–$40$. Heterogeneous nucleation on a goethite surface reduces the barrier by the wetting factor

$$f(\theta) = \frac{(2+\cos\theta)(1-\cos\theta)^2}{4}, \qquad \cos\theta = \frac{\gamma_{\mathrm{goe}} - \gamma_{\mathrm{hem|goe}}}{\gamma_{\mathrm{hem}}}. \tag{44}$$

For any hematite–goethite interfacial energy between 0.05 and 0.4 J m^{-2}, this still gives ∼240–390 k_BT under vacuum and ∼390–2400 k_BT at saturation. The observed vacuum kinetics are therefore incompatible with nucleation on free surfaces. Hematite must nucleate internally, at a low-energy goethite–hematite interface, consistent with the known topotactic dehydroxylation of goethite. In what follows, γ is replaced by an effective interfacial energy γ_{eff}. It absorbs the geometry of heterogeneous nucleation and is a property of the sample.

Calibrating γ_{eff} from the trap-isolated steps. Isolating the trap raised $P_{\mathrm{H_2O}}$ in the line and lowered $\Delta\mu$, and the isothermal rate fell by a factor s. With A and E_a unchanged, Eq. (43) gives

$$\ln\frac{1}{s} = \frac{C\gamma_{\mathrm{eff}}^3}{T}\left(\frac{1}{\Delta\mu_{\mathrm{off}}^2} - \frac{1}{\Delta\mu_{\mathrm{on}}^2}\right) \quad\Longrightarrow\quad \gamma_{\mathrm{eff}} = \left[\frac{T\ln(1/s)}{C\left(\Delta\mu_{\mathrm{off}}^{-2} - \Delta\mu_{\mathrm{on}}^{-2}\right)}\right]^{1/3}, \tag{45}$$

where $\Delta\mu_{\mathrm{on}}$ is evaluated at 10^{-8} atm and $\Delta\mu_{\mathrm{off}}$ at the water pressure reached with the trap isolated. We bound that pressure by assuming all evolved water was in the vapor, distributed uniformly through an apparatus of volume V_{app} at 298 K:

$$P_{\mathrm{H_2O}} = \frac{(m_{\mathrm{w}}/M_{\mathrm{w}})\,RT}{V_{\mathrm{app}}}. \tag{46}$$

With $m_{\mathrm{w}} = 0.027$ mg at 230 °C and 0.016 mg at 240 °C (1.50 and 0.89 μmol), and $190 \le V_{\mathrm{app}} \le 600$ mL, Table S8 gives the resulting pressures and γ_{eff}. The pressures are end-of-step values; we also evaluate half of each as a time average for steady accumulation. Because P enters only through $\ln P$ in $\Delta\mu$ and then through a cube root, a threefold uncertainty in volume changes γ_{eff} by only about 10%. The two steps agree within 15%, and we adopt

$$\gamma_{\mathrm{eff}} = 0.16 \pm 0.03 \text{ J m}^{-2}. \tag{47}$$

The corresponding barriers at 250 °C are 0.5–1.3 k_BT under vacuum and 52–129 k_BT at saturation. Under vacuum, the barrier is negligible and the

Arrhenius term of Eq. (38) sets the rate. At saturation, the barrier dominates.

Table S8: Water pressure with the trap isolated and the calibrated γ_{eff} ($\mathrm{J\,m^{-2}}$; Eq. (45)).

T (°C)	s	V_{app} (mL)	$P_{\mathrm{H_2O}}$ (Pa)	γ_{eff}, end-of-step P	γ_{eff}, $P/2$
230	0.333	190	19.6	0.138	0.145
230	0.333	600	6.2	0.150	0.158
240	0.224	190	11.6	0.164	0.172
240	0.224	600	3.7	0.178	0.188

Translation to water saturation. For the same sample, with the same internal interface and the same mobility term, only $\Delta\mu$ changes between vacuum and saturation:

$$\frac{k_{\mathrm{sat}}}{k_{\mathrm{vac}}} = \exp\left[-\frac{C\gamma_{\mathrm{eff}}^3}{T}\left(\frac{1}{\Delta\mu_{\mathrm{sat}}^2} - \frac{1}{\Delta\mu_{\mathrm{vac}}^2}\right)\right]. \tag{48}$$

For $\gamma_{\mathrm{eff}} = 0.14$, 0.16 and 0.19 $\mathrm{J\,m^{-2}}$, this ratio is 7×10^{-23}, 9×10^{-34} and 5×10^{-56} at 250 °C. At 200 °C it is 5×10^{-43}, 7×10^{-64} and 2×10^{-106}. Applying Eq. (48) step by step to the hydrothermal schedule, with k_{vac} from Eq. (38), the predicted conversion is

$$\alpha \approx \sum_i k_{\mathrm{vac}}(T_i)\,\frac{k_{\mathrm{sat}}}{k_{\mathrm{vac}}}(T_i)\,\Delta t_i = 10^{-20.6},\ 10^{-31.5},\ 10^{-53.7} \tag{49}$$

for the three values of γ_{eff}. Even the most permissive value, $\gamma_{\mathrm{eff}} = 0.084\ \mathrm{J\,m^{-2}}$, gives $\alpha = 6\times10^{-4}$. That value would require 0.1 bar of water in the line, far above the volume constraint. If the observed release came from conversion, it would imply $\alpha = F/(1-\phi) = 0.147/0.888 = 0.17$.

Figure S5 shows the difference between each observed step rate constant (Eq. (36)) and the rate predicted for its setting:

$$\Delta\log_{10} k_i = \log_{10} k_i^{\mathrm{obs}} - \log_{10} k^{\mathrm{pred}}(T_i),$$
$$k^{\mathrm{pred}} = \begin{cases} k_{\mathrm{vac}}(T) & \text{vacuum,} \\ k_{\mathrm{vac}}(T)\,\dfrac{k_{\mathrm{sat}}}{k_{\mathrm{vac}}}(T;\gamma_{\mathrm{eff}}) & \text{water-saturated.} \end{cases} \tag{50}$$

The vacuum steps in the fitting window scatter about zero with an rms of 0.15 (KAF) and 0.21 (HDM) orders of magnitude. The trap-isolated steps

sit 0.30 and 0.55 orders below. The hydrothermal steps sit 30–62 orders of magnitude above the saturated prediction for $\gamma_{\text{eff}} = 0.16$ J m^{-2}, and 20–41 and 53–105 orders above for 0.14 and 0.19 J m^{-2}. Far more ^{3}He is released in water than decomposition, as constrained by the vacuum experiments, could supply. The hydrothermal release must be controlled by diffusion.

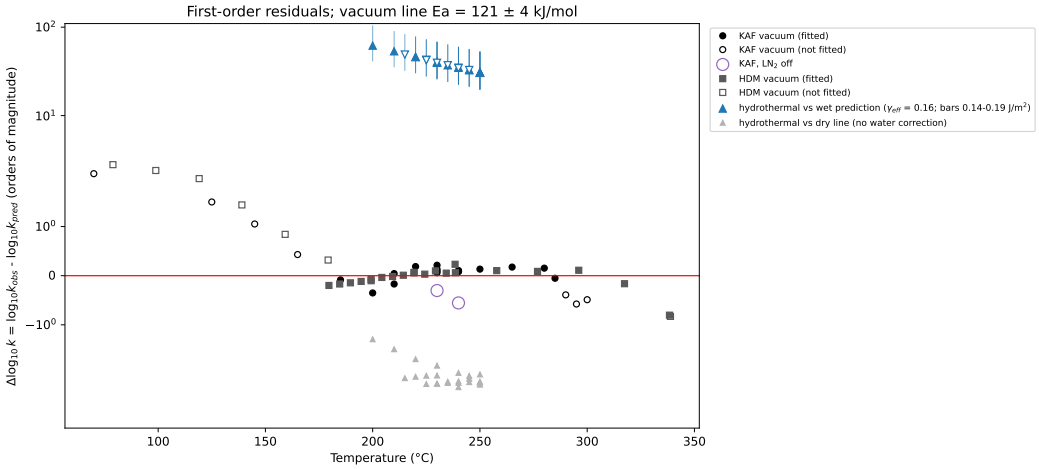


Figure S5: First-order residuals, $\log_{10} k^{\text{obs}} - \log_{10} k^{\text{pred}}$, on a symmetric-log axis. Vacuum steps (KAF, HDM) are referenced to the vacuum line (Eq. (38)); filled symbols are in the fitting window. Blue: hydrothermal steps (bounding-method central values, low-signal steps excluded) referenced to the water-saturated prediction (Eq. (48)) for $\gamma_{\text{eff}} = 0.16$ J m^{-2}; bars span 0.14–0.19 J m^{-2}. Grey: the same hydrothermal steps referenced to the vacuum line without correction for water.

A rate law linear in the driving force cannot produce this contrast. Such a law has $k \propto 1 - \exp(-\Delta\mu/RT)$, which at saturation is 0.80, 0.87 and 0.91 of the vacuum value at 150, 200 and 250 °C. That predicts only a 10–20% slowdown, far less than the trap-isolated steps and hydrothermal data show. Suppression requires a barrier that grows as $\Delta\mu^{-2}$.

This treatment is a consistent physical explanation of the data, not a proof. It rests on three assumptions. (i) The slowdown during the trap-isolated steps acts through $\Delta\mu$. If adsorbed water instead slowed dehydroxylation by blocking reactive sites, these steps would not calibrate γ_{eff}. (ii) The effective interface energy, A and E_a are the same in vacuum and in water. The mechanism is internal nucleation in both settings, and only the chemical potential of water changes. (iii) Only solid-state conversion is modeled. Conversion by dissolution of goethite and precipitation of hematite from the

fluid is not described by Eq. (48) (§8). The comparison is valid only between experimental conditions on the same sample. The calibrated γ_{eff} absorbs the sample's microstructure and should not be transferred to other goethites.

7.1. Nucleation barrier ratio and the calibration of γ_{eff}

From Eq. (42), the ratio of barriers at water saturation and under vacuum at the same temperature is

$$\frac{\Delta G^{*}_{\mathrm{sat}}}{\Delta G^{*}_{\mathrm{vac}}} = \left(\frac{\Delta\mu_{\mathrm{vac}}}{\Delta\mu_{\mathrm{sat}}}\right)^{2} \left(\frac{\gamma_{\mathrm{sat}}}{\gamma_{\mathrm{vac}}}\right)^{3}. \tag{51}$$

The prefactors, the temperature and the molar volume cancel. The first factor follows from the thermodynamic data alone (Table S2; 99 at 250 °C for bulk goethite). The second is 1 if nucleation occurs at the same internal interface in both settings. It is $(0.75/1.90)^3 = 1/16$ in the extreme case that the relevant interface is a free hematite surface, hydrated at saturation and anhydrous under vacuum. Figure S6 shows the ratio between these bounds for bulk and 10 nm goethite. Even at the lower bound, the saturated barrier is about six times the vacuum barrier at 250 °C. Because the barrier appears in an exponent, that is still a difference of many orders of magnitude in rate.

Figure S7a shows γ_{eff} from Eq. (45) as a function of the assumed water pressure during the trap-isolated steps. It spans 0.08–0.20 J m^{-2} over 10^{-5}–10^{-1} bar and 0.14–0.19 J m^{-2} over the volume-constrained range (Table S8). Figure S7b shows the predicted slowdown at saturation (Eq. (48)) for the calibrated values and for the most permissive value.

8. Hydrothermal treatment of synthetic goethite pigment

A qualitative test of fluid-mediated conversion was carried out on a commercial yellow iron-oxide pigment composed of acicular goethite crystals ∼1 µm long. These experiments were intended to give a rapid, qualitative answer, and many experimental details (exact temperature histories, run durations, and water-to-solid ratios for individual ampoules) were not recorded. At least five large silica glass ampoules were each loaded with pigment and an excess of water and fused shut at ambient pressure. They were prepared far less carefully than the step-heat ampoule of main text §2.3; their larger size made them much easier to handle. The ampoules were heated in an oven without temperature control: the temperature was monitored and the

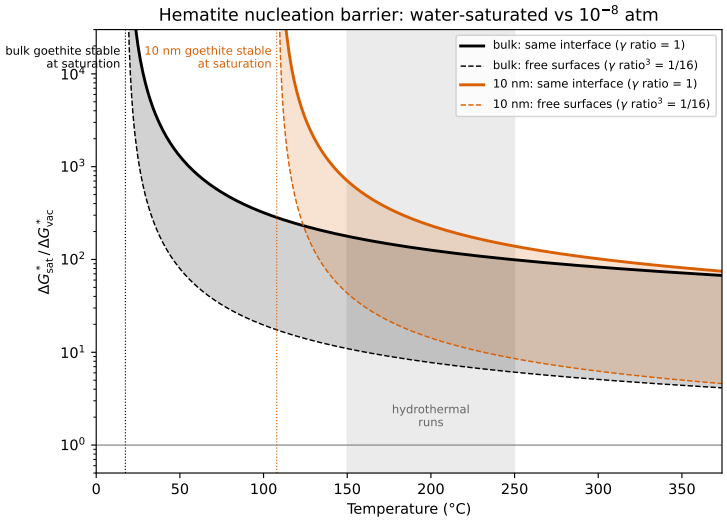


Figure S6: Ratio of hematite nucleation barriers at water saturation and at $P_{\mathrm{H_2O}} = 10^{-8}$ atm for bulk and 10 nm goethite. Solid: same interface in both settings; dashed: free hematite surfaces (hydrated vs anhydrous). Below the vertical dotted lines, goethite is stable at saturation and there is no driving force.

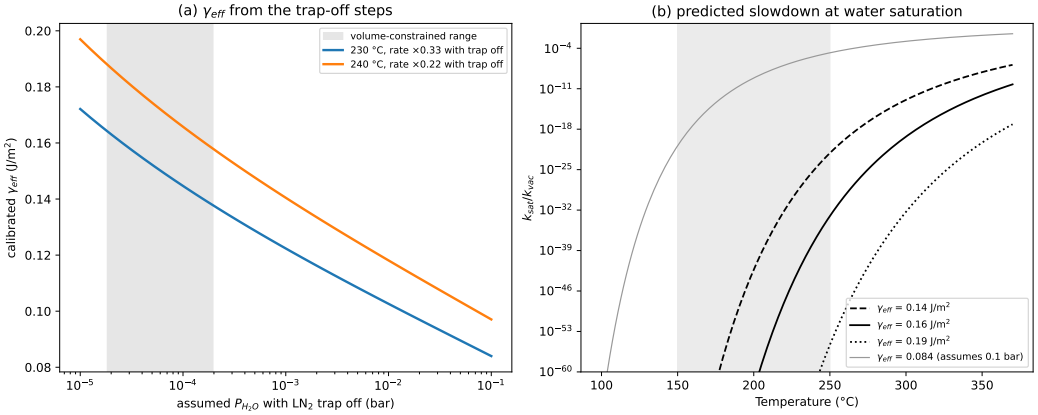
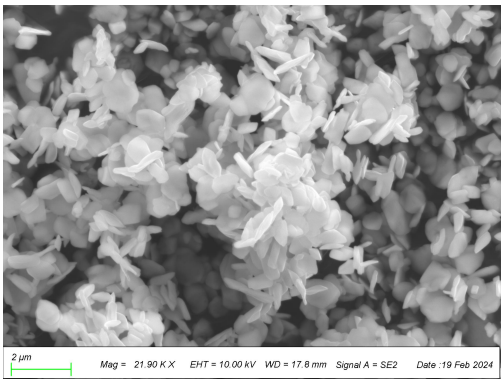


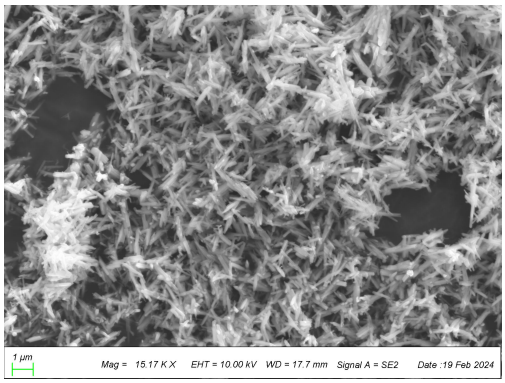
Figure S7: (a) γ_{eff} calibrated from the two trap-isolated KAF steps as a function of the assumed $P_{\mathrm{H_2O}}$; grey band: range allowed by the apparatus volume (190–600 mL) and the water released in each step. (b) Ratio of conversion rate constants at water saturation and under vacuum (Eq. (48)); grey band: temperatures of the hydrothermal experiments.

heater voltage adjusted by hand to keep it between 200 and 250 °C, and the ampoules were held for hours, typically about a day. Reaction was assessed by color (yellow goethite, red hematite), and representative unreacted and reacted products were imaged in secondary electron mode (as in main text §2.5).

Fluid-mediated conversion does occur in some materials. In the qualitative experiments on acicular synthetic goethite pigment described above, heated with excess water at 200–250 °C for about a day, the outcome was all or nothing. Roughly equal proportions of ampoules were completely reacted (red), completely unreacted (yellow), or contained both reacted and unreacted material. In the mixed ampoules, the red and yellow portions were entirely isolated from one another, separated by gas bubbles. Unreacted pigment retains its acicular habit (Fig. S8b). The reacted product is instead composed of equant to platy crystals ∼0.5–1 µm across (Fig. S8a), interpreted as hematite from its color and habit. The change of shape rules out pseudomorphic dehydroxylation. It indicates dissolution of goethite and nucleation and growth of hematite from the fluid, a pathway outside the solid-state model of §7.



(a) Reacted (red) product



(b) Unreacted (yellow) pigment

Figure S8: Secondary electron images of synthetic goethite pigment after hydrothermal treatment at 200–250 °C for about a day. (a) Material from a completely reacted (red) ampoule: the acicular goethite has been replaced by equant to platy crystals ∼0.5–1 µm across, interpreted as hematite. (b) Material from an unreacted (yellow) ampoule, which retains the acicular habit of the starting pigment. The change in crystal habit requires dissolution of goethite and precipitation of hematite from the fluid, and the all-or-nothing outcome of the experiments indicates that the reaction is limited by nucleation and proceeds rapidly once it begins.

The distribution of outcomes is itself diagnostic. Conversion either went to completion within a fluid-connected volume or did not occur at all, even where reacted and unreacted pigment shared the same ampoule and thermal history. This is the behavior of a nucleation-limited process: once hematite nucleates, the reaction proceeds rapidly to completion, and where it does not nucleate, goethite persists for the full run. The fluid-mediated pathway is therefore governed, like the solid-state pathway, by the difficulty of forming a hematite nucleus rather than by the rate at which hematite grows.

Grain size and bulk thermodynamics do not explain why these blades reacted when the natural goethite did not. At 1 µm, the surface contribution to ΔG_r is only $\sim$0.04 kJ, and the bulk driving force at saturation is the same 6–11 kJ for both materials (Table S2). The most likely difference is the fluid itself. Commercial iron-oxide pigments are commonly produced by precipitation from iron-salt solutions. They can carry residual soluble salts, such as sulfate, and may be surface-treated. Such contaminants would change the pH and ionic strength of the water they are heated in, and with them the solubility of Fe(III). The dissolution–precipitation pathway is limited by how much iron the fluid can carry, so a modest increase in Fe(III) solubility could turn a negligible reaction into a visible one. Differences in residue loading or in the handling of individual ampoules would also explain why the outcome was not reproducible. The natural goethite, by contrast, was heated in distilled water, where near-neutral Fe(III) solubility is very low.

We conclude that water-saturated conditions suppress solid-state conversion of goethite, but they do not by themselves exclude fluid-mediated conversion. Whether that pathway operates depends on the solution chemistry. For the natural sample in distilled water, §6 shows that the release is diffusive; a nucleation-limited, all-or-nothing reaction would also be expected to release helium abruptly rather than in the smooth, history-dependent manner observed. For samples whose chemistry is less well characterized, the absence of conversion should be verified directly, for example by post-run X-ray diffraction or electron microscopy.

References

Bell, I.H., Wronski, J., Quoilin, S., Lemort, V., 2014. Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp. Industrial & Engineering Chemistry Research 53, 2498–2508. doi:10.1021/ie4033999.

Berman, R.G., Brown, T.H., 1985. Heat capacity of minerals in the system Na2O-K2O-CaO-MgO-FeO-Fe2O3-Al2O3-SiO2-TiO2-H2O-CO2: representation, estimation, and high temperature extrapolation. Contributions to Mineralogy and Petrology 89, 168–183.

Carslaw, H.S., Jaeger, J.C., 1959. Conduction of Heat in Solids. 2nd ed., Oxford University Press, Oxford.

Farley, K.A., Monteiro, H.B., Vasconcelos, P.M., Waltenberg, K., 2024. Dehydration of goethite during vacuum step-heating and implications for he retentivity characterization. Chemical Geology 663, 122254.

Farley, K.A., Wolf, R.A., Silver, L.T., 1996. The effects of long alpha-stopping distances on (u-th)/he ages. Geochimica et Cosmochimica Acta 60, 4223–4229.

Fechtig, H., Kalbitzer, S., 1966. The diffusion of argon in potassium-bearing solids, in: Schaeffer, O.A., Zähringer, J. (Eds.), Potassium Argon Dating. Springer, Berlin, Heidelberg, pp. 68–107.

Goldsmith, A.S., Ketcham, R.A., Stockli, D.F., 2020. Simulating effects of heterogeneous 4he concentration profiles and radiation damage annealing on whole-grain zircon diffusivity analyses. Geochimica et Cosmochimica Acta 284, 239–253.

Jones, W.M., 1953. Permeability and solubility of He^3 and He^4 in vitreous silica. Journal of the American Chemical Society 75, 3093–3096.

Ketcham, R.A., Gautheron, C., Tassan-Got, L., 2011. Accounting for long alpha-particle stopping distances in (u-th-sm)/he geochronology: Refinement of the baseline case. Geochimica et Cosmochimica Acta 75, 7779–7791.

Majzlan, J., Grevel, K.D., Navrotsky, A., 2003a. Thermodynamics of Fe oxides: Part II. enthalpies of formation and relative stability of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3). American Mineralogist 88, 855–859.

Majzlan, J., Lang, B.E., Stevens, R., Navrotsky, A., Woodfield, B.F., Boerio-Goates, J., 2003b. Thermodynamics of Fe oxides: Part I. entropy at standard temperature and pressure and heat capacity of goethite (α-FeOOH),

lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3). American Mineralogist 88, 846–854.

Mampel, K.L., 1940. Zeitumsatzformeln für heterogene Reaktionen an Phasengrenzen fester Körper. 2. Die Zeitumsatzformeln für ein Pulver aus kugelförmigen Teilchen. Zeitschrift für Physikalische Chemie 187A, 235–249. doi:10.1515/zpch-1940-18727.

Monteiro, H., 2017. Paleoenvironmental Evolution of Continental Landscapes through Combined High-Resolution Geochronology and δ^{18}O Ion Microprobe Analysis of Goethite. Ph.D. thesis. The University of Queensland. doi:10.14264/uql.2018.6.

Navrotsky, A., Mazeina, L., Majzlan, J., 2008. Size-driven structural and thermodynamic complexity in iron oxides. Science 319, 1635–1638.

Shuster, D.L., Farley, K.A., Sisterson, J.M., Burnett, D.S., 2004. Quantifying the diffusion kinetics and spatial distributions of radiogenic 4he in minerals containing proton-induced 3he. Earth and Planetary Science Letters 217, 19–32.

Swets, D.E., Lee, R.W., Frank, R.C., 1961. Diffusion coefficients of helium in fused quartz. The Journal of Chemical Physics 34, 17–22.

Voigt, D.E., Will, G., 1981. TODO: verify title. Neues Jahrbuch für Mineralogie, Monatshefte 2, 89.

Wagner, W., Pruß, A., 2002. The IAPWS formulation 1995 for the thermodynamic properties of ordinary water substance for general and scientific use. Journal of Physical and Chemical Reference Data 31, 387–535. doi:10.1063/1.1461829.

Wilson, W.D., Bisson, C.L., 1973. Atomistics of helium diffusion in copper and tungsten. Radiation Effects 19, 53–58.

Yakubovich, O.V., Shukolyukov, Y.A., Kotov, A.B., Yakovleva, S.Z., Sal'nikova, E.B., 2010. Geothermochronology based on noble gases: II. stability of the (u-th)/he isotope system in zircon. Petrology 18, 555–570.